Problem 2

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Problem 1 *The velocity of an object moving along a straight line is given by $v(t)$ (in ft/s) and we only know the following:* $\int_0^3 v(t)\,dt = -3$, *and* $\int_3^4 v(t)\,dt = 5$.

Compute the following displacements, if possible. If it is not possible, give examples explaining why not.

(a) $s(3) - s(0)$

Explanation.

$$s(3) - s(0) = \int_0^3 v(t)\,dt = -3ft$$

(b) $s(4) - s(0)$

Explanation. First notice that

$$s(4) - s(0) = \int_0^4 v(t)\,dt$$

Therefore,

$$s(4) - s(0) = \int_0^3 v(t)\,dt + \int_3^4 v(t)\,dt = -3 + 5 = 2ft$$

(c) *Find the displacement during the interval* $[0, 4]$ *if the velocity at the time* t, $0 \le t \le 4$, *was* $v(t) + 2$ *ft/s instead?*

Explanation. In that case, we would have that

$s(4) - s(0) = \int_0^4 (v(t) + 2)\,dt = \int_0^4 v(t)\,dt + \int_0^4 2\,dt$

We use the result in part (b) for the first integral, and geometry for the second interval.

$s(4) - s(0) = 2 + 2(4 - 0) = 2 + 8 = 10$ ft

(d) *Find the displacement during the interval* $[0, 4]$ *if the velocity at the time* t, $0 \le t \le 4$, *was* $5v(t)$ *ft/s instead?*

Explanation. $s(4) - s(0) = \int_0^4 5v(t)\,dt = 5\int_0^4 v(t)\,dt = 5(2) = 10$ ft